

Bobcat Pin List and Electrical Specifications

1. Pin Summary

Pin #	Name	Type	Direction	Description
1	VDDA1	Power	Supply	Analog core power
26,27	VDDA2	Power	Supply	Analog core power
12,20	VDDD	Power	Supply	Digital core power
7,13,22,33,34	VDDIO	Power	Supply	IO power
41	GND	Power	Ground	Analog, Digital, and IO ground
14	MOSI	Digital	Input	SPI data in (Master Out Slave In)
15	MISO	Digital	Output	SPI data out (Master In Slave Out)
16	SCLK	Digital	Input	SPI clock (idle low)
17	CS_L	Digital	Input	SPI chip select (active low)
18	SPI_DMODE	Digital	Input	SPI data mode select
40	GPIO0	Digital	Bidirectional	General-purpose IO
39	GPIO1	Digital	Bidirectional	General-purpose IO
38	GPIO2	Digital	Bidirectional	General-purpose IO
37	GPIO3	Digital	Bidirectional	General-purpose IO
2	SAMPLE_OUTV	Digital	Output	Sample data output valid
3	SAMPLE_OUT0	Digital	Output	Sample data output bit 0
4	SAMPLE_OUT1	Digital	Output	Sample data output bit 1
5	SAMPLE_OUT2	Digital	Output	Sample data output bit 2
6	SAMPLE_OUT3	Digital	Output	Sample data output bit 3
7	SAMPLE_OUT4	Digital	Output	Sample data output bit 4
8	SAMPLE_OUT5	Digital	Output	Sample data output bit 5
9	SAMPLE_OUT6	Digital	Output	Sample data output bit 6
10	SAMPLE_OUT7	Digital	Output	Sample data output bit 7
36	CLK_OUT0	Digital	Output	Oscillator clock output
35	CLK_OUT1	Digital	Output	Oscillator clock output
32	CLK_OUT2	Digital	Output	Oscillator clock output
31	CLK_OUT3	Digital	Output	Oscillator clock output
19	RESET_N	Digital	Input	Chip reset (active low)
23	OSC_EN	Digital	Input	Oscillator enable
24	WEIGHT_EN	Digital	Input	Weight enable
25	SAMPLE_TRIG	Digital	Input	Sample trigger
28	BIAS0	Analog	Input	Bias current (320uA, 0.5V nominal)
29	BIAS1	Analog	Input	Bias current (320uA, 0.5V nominal)

2. Power Supply Specifications

Supply	Min	Typ	Max	Unit
VDDD	0.72	0.80	0.88	V
VDDA1	0.72	0.80	0.88	V
VDDA2	0.72	0.80	0.88	V
VDDIO	1.62	1.80	1.98	V

3. Digital IO Electrical Characteristics

Parameter	Symbol	Min	Typ	Max	Unit
Input high voltage	VIH	$0.65 \times VDDIO$	--	$VDDIO + 0.3$	V
Input low voltage	VIL	-0.3	--	$0.35 \times VDDIO$	V
Output high voltage	VOH	$VDDIO - 0.45$	--	--	V
Output low voltage	VOL	--	--	0.45	V